

Daily Research Digest — 2026-04-10

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

Executive overview

Recent preprints across various astrophysical and cosmological topics reveal an ongoing exploration of dark matter, particularly its interaction with baryonic components and gravitational lensing. Additionally, new insights into galaxy formation and evolution are emerging from detailed studies of galactic structures and their environmental dependencies.

Galaxy clusters

This section is currently lacking recent preprints for direct synthesis. However, ongoing research continues to explore the impact of dense environments on galaxy properties within clusters.

Galaxies

Recent work highlights a multi-faceted approach towards understanding galaxy morphology and evolution through detailed analysis. The study of NGC 628 showcases the use of ordinal patterns and statistical measures like permutation entropy (H), disequilibrium (D_E), and complexity (C) to reveal spatial scales transitioning from star formation to larger-scale dynamics (1). Additionally, the investigation into neutral hydrogen content in the Shapley Supercluster core emphasizes environmental effects on gas content and star-forming activity across different galaxy populations, noting a significant prevalence of transition galaxies within this dense environment compared to field samples (2).

Gravitational lensing

Gravitational lensing research is advancing with innovative modeling techniques. One paper employs ray-traced weak gravitational lensing to probe ultralight scalar dark matter around S-stars in the Galactic Center, utilizing their orbital dynamics to constrain ULDM parameters (3). Another study delves into Majoron Dark Matter through its interaction with photons in optical interferometers, potentially detectable via birefringence effects caused by a coherent dark matter background (4).

Dark Matter

Recent investigations into dark matter have introduced novel scenarios and constraints. The exploration of multi-component dark matter models with singlet fermions and scalar particles highlights the complexity of dark matter interactions while probing viable regions in parameter space that align with observed relic density (5). Additionally, the study of ultralight scalar dark matter leverages gravitational lensing effects on S-stars to impose stringent constraints on ULDM mass ratios relative to the supermassive black hole at the Galactic Center (6).

Constraining Ultralight Scalar Dark Matter in the Galactic Center with the S2 Orbit

The dense environment around the supermassive black hole, Sgr A*, provides a unique setting for probing ultralight dark matter through its impact on the orbital dynamics of stars like S2. This research constrains ULDM mass and coupling constants using the periastron precession rate observed in S2's orbit (7).

Probing Majoron Dark Matter with Gravitational Wave Detectors

An interesting approach to detecting Majoron dark matter involves its interaction with photons via a QED anomaly, leading to birefringence detectable by optical interferometers like Advanced LIGO and KAGRA. This research proposes new methods for probing the coupling between Majoron dark matter and Standard Model particles (8).

Multi Component Dark Matter in a Minimal Model

Research on multi-component dark matter incorporates singlet fermions and scalars to explore viable regions of parameter space that match observed relic densities while considering loop-suppressed DM-nucleon scattering cross sections (9).

Too Big to Quench? I. Constraining ISM Stripping of Dwarf Satellites in Milky Way-like Halos

This study uses high-resolution hydrodynamical simulations to constrain the efficiency of interstellar medium (ISM) stripping in dwarf satellite galaxies within Milky Way-like halos, highlighting mass scales at which star formation quenching becomes inefficient due to ram pressure stripping (10).

Conclusions and Future Directions

The preprints underscore a diverse range of methodologies and theoretical frameworks being employed to understand dark matter properties and interactions with baryonic components. Detailed analysis of galactic structures and their environmental dependencies continues to shed light on the complex processes governing galaxy formation and evolution. Ongoing research promises further insights through innovative observational techniques and simulation studies.

References

1. Ordinal Patterns Analysis in NGC 628
2. Neutral Hydrogen Content in Shapley Supercluster Core
3. Constrained Ultralight Scalar Dark Matter with S2 Orbit
4. Majoron Dark Matter Probed by Birefringence Effects
5. Multi-Component Dark Matter Minimal Model Study
6. Gravitational Lensing Constraints on ULDM Mass Ratios
7. Orbital Dynamics of S2 to Probe ULDM in Galactic Center
8. QED Anomaly and Photon-Birefringence for Majoron Detection
9. Singlet Fermions and Scalars in Multi-Component DM Models
10. ISM Stripping Efficiency in Dwarf Satellites within Milky Way-like Halos

Paper Summaries and Figures

1. Coupled Dark Energy and Dark Matter for DESI: An Effective Guide to the Phantom Divide

Focus: dark matter

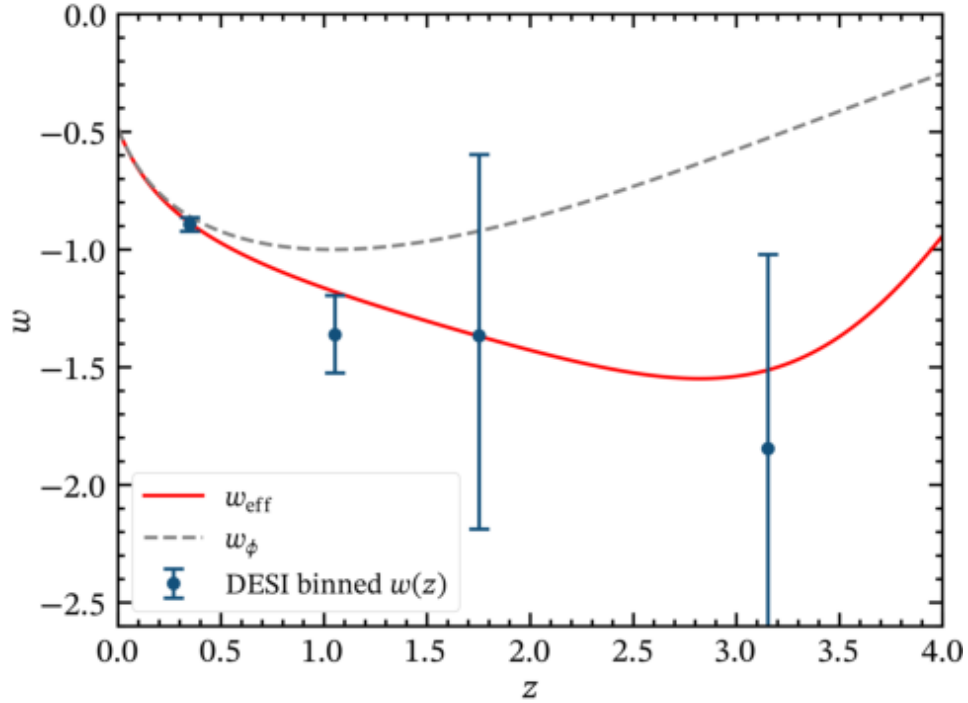
Authors: Stefan Antusch, Stephen F. King, Xin Wang

Published: 2026-04-09T16:45:31+00:00

PDF: <https://arxiv.org/pdf/2604.08449v1>

Summary: Motivated by the recent Dark Energy Spectroscopic Instrument (DESI) DR2 preference for dynamical dark energy, we study interacting dark energy models in which a canonical quintessence field couples to cold dark matter through a field-dependent mass $m(\phi)$. In such scenarios, the effective equation of state inferred under the assumption of non-interacting dark sectors, $w_{\text{eff}}(z)$, can differ from the intrinsic scalar-field equation of state $w_\phi(z)$, making an apparent phantom crossing $w_{\text{eff}} < -1$ possible without introducing a phantom scalar. We show that a viable realization of this mechanism requires the scalar field to originate from a frozen phase deep in the radiation era, in order for the effective coupling to remain sufficiently suppressed before recombination to evade cosmic microwave background constraints, and for the late-time evolution to become strong enough to reproduce the apparent behavior of $w_{\text{eff}}(z)$ preferred by DESI. We identify the general conditions that allow these requirements to be satisfied simultaneously, and present an illustrative phenomenological realization in which $w_{\text{eff}}(z)$ evolves from $w_{\text{eff}} \approx -1.2$ at $z \approx 1.0$ to $w_{\text{eff}} \approx -0.9$ at $z \approx 0.4$. These conditions and requirements serve as a guide for designing future models of this kind which can safely navigate the phantom divide at $w = -1$ in an effective way without phantom fields.

Selected figures



2. Optical images of Kerr-Sen black hole illuminated by thick accretion disks

Focus: gravitational lensing

Authors: Yu-Kang Wang, Chen-Yu Yang, Xiao-Xiong Zeng

Published: 2026-04-09T16:38:30+00:00

PDF: <https://arxiv.org/pdf/2604.08439v1>

Summary: This paper investigates the shadow and polarization images of a Kerr-Sen black hole illuminated by geometrically thick and optically thin accretion disks. We adopt two classes of accretion models, namely the phenomenological radiatively inefficient accretion flow (RIAF) model and the analytical ballistic approximation accretion flow (BAAF) model. Based on radiative transfer theory, we examine the effects of the spin parameter a , black hole charge Q , and observer inclination angle on the shadow images. Both models show that, as the charge Q increases, the photon rings and the central dark regions shrink simultaneously. Meanwhile, frame dragging gives rise to a pronounced brightness asymmetry, which becomes more significant with increasing a and Q . The main difference between isotropic and anisotropic radiation is that, in the latter case, the higher order images are brighter in the upper and lower polar regions. For the BAAF model, because the conical approximation renders certain regions geometrically thinner, the spatial extent of the higher order images is narrower than that in the RIAF model, and the separation between the direct image and the higher order images is more distinct. In the polarization images, the spatial distribution of the polarization vector directions is mainly determined by gravitational lensing and frame dragging, whereas the intensity near the photon ring and the scale of the higher order images are significantly influenced by Q .

3. Morphological complexity of NGC 628 - a multiwavelength multiscale analysis using the ordinal pattern framework

Focus: galaxies

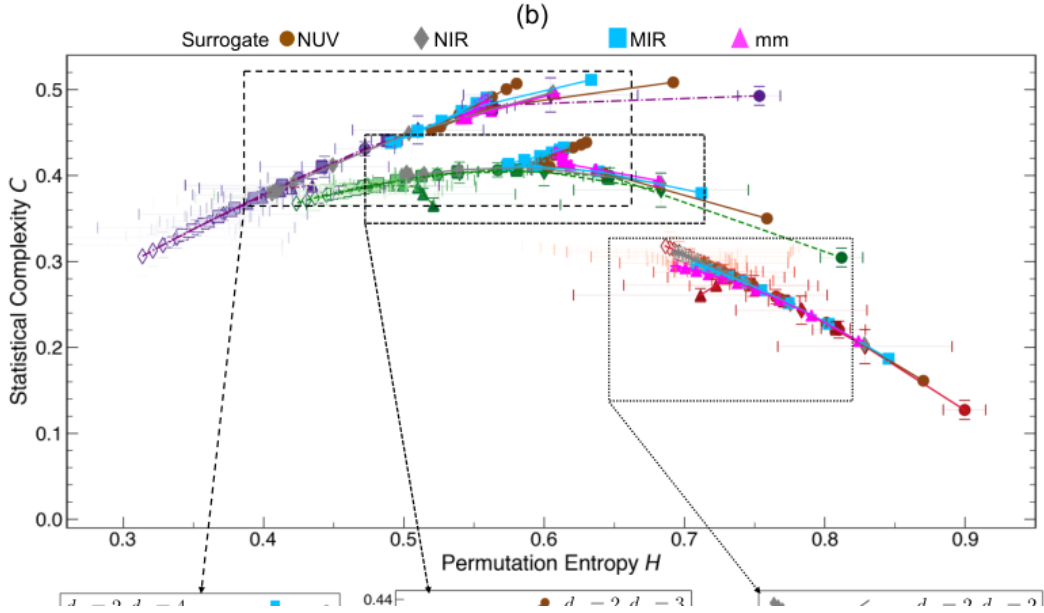
Authors: Athokpam Langlen Chanu, S Amrutha, Pravabati Chingangbam, Changbom Park

Published: 2026-04-09T16:08:41+00:00

PDF: <https://arxiv.org/pdf/2604.08409v1>

Summary: As statistical systems, galaxies exhibit a rich interplay between organized structure and stochastic fluctuations across a broad range of spatial scales. This duality motivates the need for quantitative frameworks capable of capturing their morphological complexity. The ordinal patterns framework, along with its associated statistical measures: permutation entropy (H), disequilibrium (D_E), statistical complexity (C), and ordinal network node entropy, has recently emerged as a powerful tool for analyzing such complexity in physical systems. We apply this framework in a multiwavelength, multiscale analysis of the galaxy NGC 628, utilizing observations in the near-ultraviolet, near-infrared, mid-infrared, and millimeter bands. Our results reveal a characteristic spatial scale of approximately 200 parsecs, marking the transition from small-scale structures influenced by star formation and stellar feedback to larger-scale morphology governed by the galaxy's dynamics. Furthermore, we find that the C vs. H trajectories for all wavelengths converge toward a common attractor curve, consistent with the behavior of isotropic Gaussian random fields. This convergence suggests a universal statistical behavior in galactic structure at large scales, despite the differing physical processes traced by each wavelength.

Selected figures



4. Neutral Hydrogen in the Shapley Supercluster Core I: Environmental Effects on Gas Content and Galaxy Evolution

Focus: galaxies

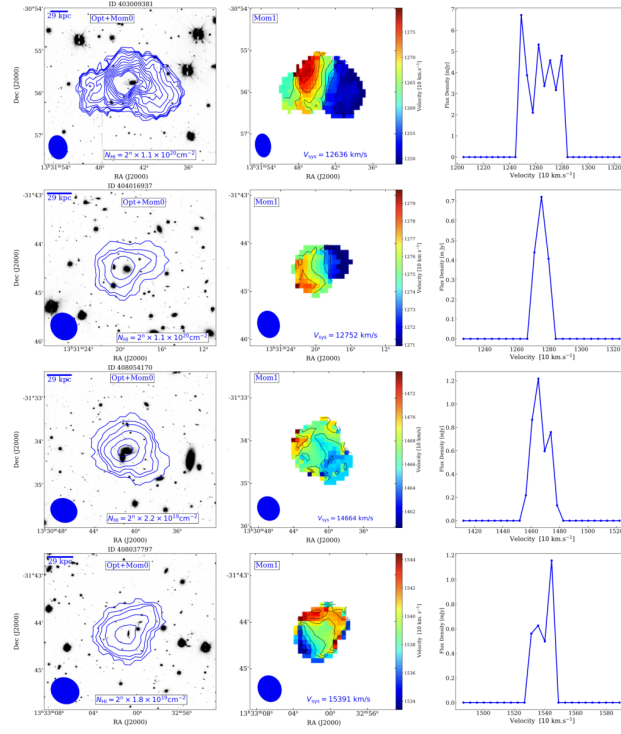
Authors: L. Gwebushe, T. Venturi, P. Merluzzi, G. Busarello, V. Casasola, O. Smirnov, M. Ramatsoku, J. Dawson

Published: 2026-04-09T16:01:32+00:00

PDF: <https://arxiv.org/pdf/2604.08402v1>

Summary: We study the atomic Hydrogen (HI) content of galaxies in the core of the Shapley Supercluster (SSC) at $\langle z \rangle \sim 0.048$, using observations from the MeerKAT Galaxy Cluster Legacy Survey and optical data from the Shapley Supercluster Survey (ShaSS) project. Our sample comprises 169 galaxies with HI detections in the dynamically active region of Abell 3558 and SC1329-313. Following the literature, we classify galaxies into star-forming main sequence (SFMS), transition (TZ), and red sequence (RS) populations, and examine how the HI content varies across these populations. Galaxies on the SFMS exhibit an average HI gas fraction offset of 0.038 dex from the gas fraction main sequence, while TZ and RS populations show depleted HI fractions of -0.034 and -0.211 dex. HI depletion timescales span from 6 to 170 Gyr (SFMS-TZ-RS) confirming increasingly inefficient star formation with quenching. Scaling relations between HI mass and stellar mass in the SSC are generally consistent with field samples. The most direct signature of the dense environment of the SSC is the marked predominance of TZ galaxies, in contrast to what is observed in the field-dominated sample of xGASS, where the population is mostly composed of SFMS galaxies. Moreover, the SFMS and RS populations have similar size, again in contrast with field populations. These results suggest that galaxies in the SSC are undergoing environmental quenching through starvation or strangulation, rather than rapid gas stripping. Despite detectable HI reservoirs, many galaxies exhibit long depletion times, indicating reduced gas accretion and inefficient star formation.

Selected figures



5. Ray-traced weak lensing convergence in screened modified gravity theories

Focus: gravitational lensing

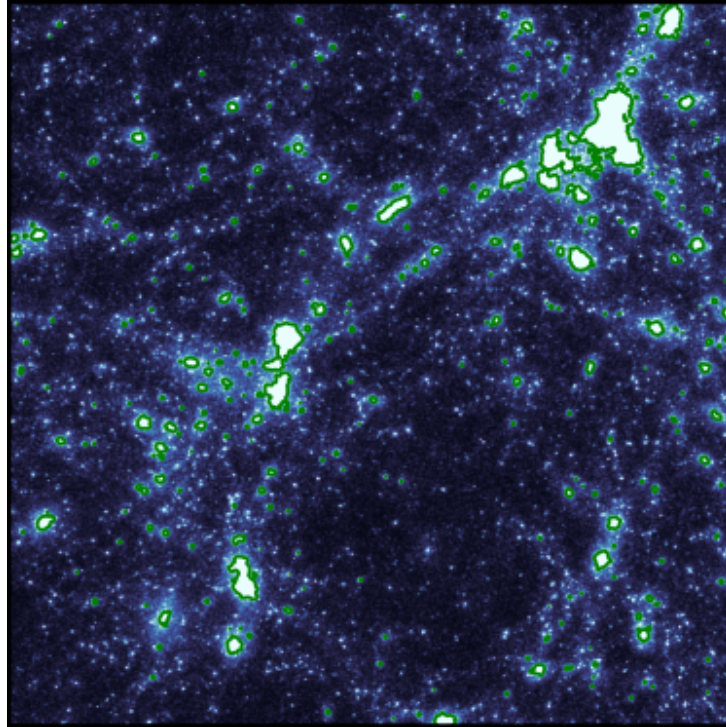
Authors: Mattia Pantiri, Matthieu Schaller, Alessandra Silvestri, Jeger C. Broxterman, Joop Schaye

Published: 2026-04-09T15:54:01+00:00

PDF: <https://arxiv.org/pdf/2604.08393v1>

Summary: Weak gravitational lensing is one of the primary cosmological probes, providing powerful constraints on the cosmological model. As Stage IV surveys are expected to deliver data of unprecedented precision, accurate modeling of weak gravitational lensing observables across both linear and non-linear scales becomes increasingly important. In this work, we investigate weak lensing in modified gravity (MG) models, extensions of the standard Λ CDM cosmology in which gravity deviates from general relativity, generally introducing modifications to the lensing equation. We parametrize these modifications through the common phenomenological function μ_{mg} and apply ray-tracing to the density maps of N-body and hydrodynamical simulations. We model the time dependence of μ_{mg} analytically, while we introduce a phenomenological scale dependence to represent the screening mechanisms by which MG models reduce to general relativity in high-density environments. Starting from the output of the FLAMINGO hydrodynamical simulations, we generate fully ray-traced convergence maps using our modified lensing model. We analyze how the parameters of our prescription affect the weak lensing convergence power spectrum and compare these effects to other known sources of variation, in particular cosmological parameters and baryonic feedback. We find that the modifications to the lensing equation deriving from the MG model produce non-negligible signatures in the convergence power spectrum and that, within extensions of the Λ CDM framework, these effects can be larger than those induced by baryonic physics. Our results indicate that modified lensing should become a standard ingredient of the analysis of modified gravity simulations.

Selected figures



6. Extended coronal line emission and new clues to a possible dual AGN in the merger J1356+1026

Focus: galaxies

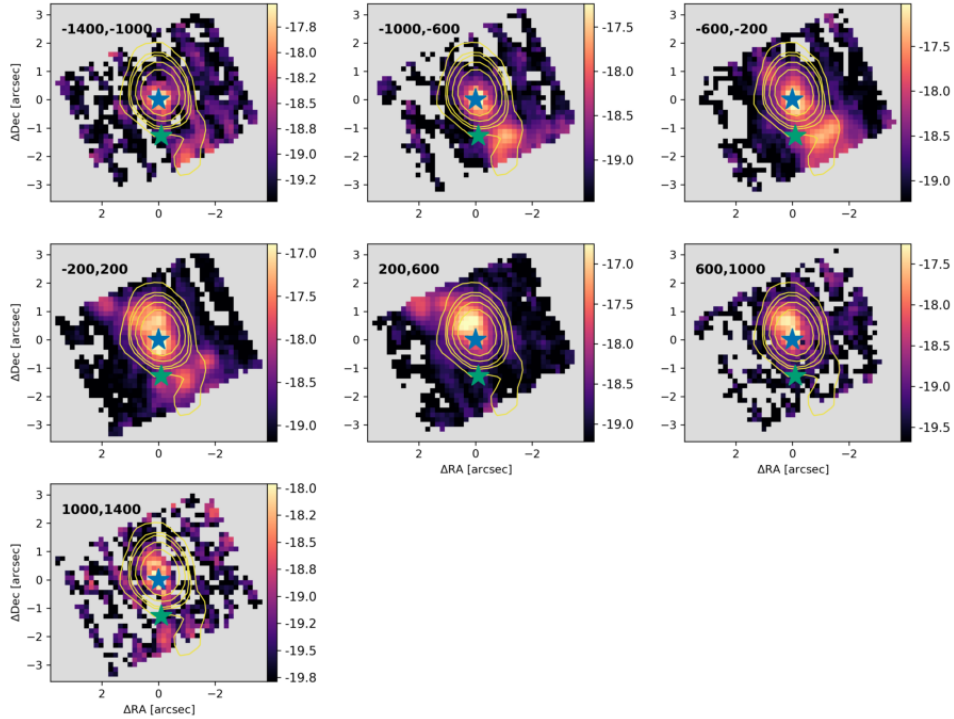
Authors: M. Bianchin, C. Ramos Almeida, O. González-Martín, M. V. Zanchettin, M. Carneiro, M. Pereira-Santaella, C. Tadhunter, G. Speranza, I. García-Bernete, A. Audibert, A. Alonso-Herrero, D. Rigopoulou, A. Labiano, J. A. Acosta-Pulido, S. García-Burillo

Published: 2026-04-09T13:30:27+00:00

PDF: <https://arxiv.org/pdf/2604.08239v1>

Summary: Merging luminous galaxies are ideal laboratories to study some of the most extreme astrophysical phenomena. The local ($z=0.1232$) obscured quasar J1356+1026 has two nuclei, North and South (J1356N and J1356S), but despite numerous efforts, J1356S had not yet been confirmed as an AGN. Thanks to the superb sensitivity and spatial resolution of the MIRI/MRS instrument on board the JWST, we present new evidence suggesting that J1356S may indeed host an AGN with $\log L_{\text{bol}} = 43.4 \pm_{0.5}^{0.6} \text{ ergs}^{-1}$. This is supported by the detection of strong coronal line emission at this location and by a spectral shape that differs from that of J1356N and those of the narrow-line region (NLR). Aided by the spatially resolved information of MIRI/MRS and VLT/SINFONI, we also find that the high ionization gas, traced by the coronal lines [Ne V] $14.3 \mu\text{m}$ and [Si VI] $1.963 \mu\text{m}$, has an extension of $\sim 13\text{--}15.5 \text{ kpc}$. This is likely a lower limit of the true extension, as suggested by the comparison with optical imaging from HST. {The extended [Ne V] emission can be accounted for by photoionization from the quasar in J1356N in a relatively low density environment, ranging from $n_e \leq 2000 - 3800 \text{ cm}^{-3}$ in J1356N and $n_e \leq 600 - 1200 \text{ cm}^{-3}$ in J1356S and the NLR, as measured from the [Ne V] $14.3 \mu\text{m}$ and $24.3 \mu\text{m}$ lines.

Selected figures



7. Probing Majoron Dark Matter with Gravitational Wave Detectors

Focus: dark matter

Authors: Ippei Obata, Tsutomu T. Yanagida

Published: 2026-04-09T12:46:40+00:00

PDF: <https://arxiv.org/pdf/2604.08193v1>

Summary: The Majoron is a hypothetical (pseudo) Nambu-Goldstone boson arising from the spontaneous breaking of a global lepton number symmetry, and is known as a candidate for dark matter in our Universe. In this paper, we investigate the possibility of probing the Majoron dark matter with a linear optical cavity used in the interferometric gravitational wave detectors. We consider a scenario in which the Majoron dark matter couples to photons through a QED anomaly, leading to an oscillatory photon birefringence induced by the coherent dark matter background. The anomaly coefficient is fixed by requiring the model to simultaneously reproduce the electroweak Higgs scale and a typical right-handed Majorana neutrino mass scale, and the resulting dark matter-photon coupling naturally falls within the sensitivity range of optical interferometers. By incorporating additional optics to extract the birefringence signal, we find that ground-based laser interferometers such as Advanced LIGO, KAGRA, as well as future detectors, can probe a region of the parameter space of Majoron dark matter.

8. Constraining Ultralight Scalar Dark Matter in the Galactic Center with the S2 Orbit

Focus: dark matter

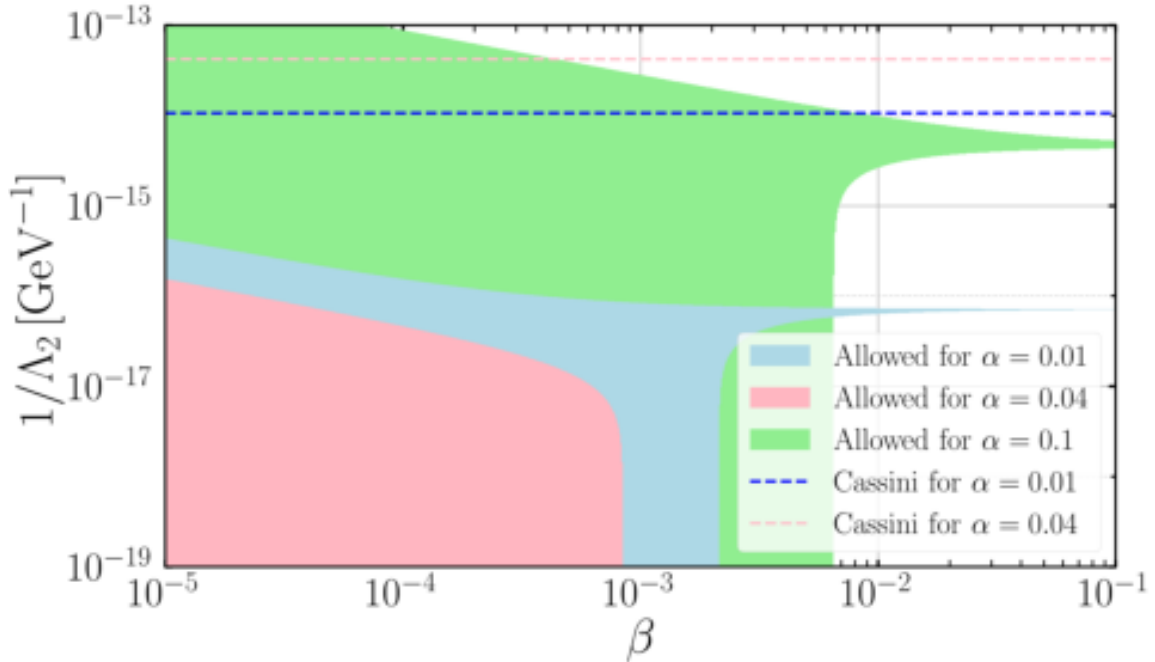
Authors: Jiang-Chuan Yu, Yan Cao, Lijing Shao

Published: 2026-04-09T10:02:09+00:00

PDF: <https://arxiv.org/pdf/2604.08053v1>

Summary: The dense environment of our Galactic Center (GC) offers a unique laboratory for probing ultralight dark matter (ULDM). We explore the prospect of detecting a scalar ULDM field through its effects on the orbital dynamics of S-stars around the supermassive black hole in the GC, Sgr A*. We consider both linear and quadratic couplings between the real scalar field ϕ and Standard Model particles, and analyze two representative ULDM structures: the scalar gravitational atom and the spherical soliton. We find that quadratic coupling induces a non-oscillatory perturbation, leading to a long-term secular orbital evolution. We use the observed periastron precession rate of S2 star to put stringent constraints on the total ULDM mass in the GC and the quadratic coupling constant. For the gravitational atom $|211\rangle$ state, we constrain the mass ratio of ULDM to Sgr A* to $\beta \lesssim 10^{-3}$ at $m \sim 10^{-18}$ eV, and for the spherical soliton which extends to ~ 0.2 pc, the mass ratio is limited to $\beta \lesssim 1$ at $m \sim 3 \times 10^{-20}$ eV. Notably, the resulting limits on the quadratic coupling constant surpass current bounds in the mass range 10^{-20} eV $\lesssim m \lesssim 10^{-18}$ eV.

Selected figures



9. Multi Component Dark Matter in a Minimal Model

Focus: dark matter

Authors: Karim Ghorbani

Published: 2026-04-08T21:39:30+00:00

PDF: <https://arxiv.org/pdf/2604.07618v1>

Summary: We study a minimal model with an imposed $\mathbb{Z}_2$ symmetry incorporating two singlet fermions and a singlet scalar which communicate with the SM particles through a scalar-Higgs portal. We probe regions in the parameter space where stability of the three new particles are guaranteed kinematically, and thus introducing a multi component dark matter (DM) scenario. The regions in the parameter space with predicted total relic density in accordance with the observed DM abundance are found, and the contribution of each species to the total relic density is determined. The elastic DM-nucleon scattering cross section of the two fermions DM are loop suppressed, while that of the scalar DM starts at tree level and thus presumably being dominant. It is found that there exist a viable region in the parameter space that the scalar DM can evade the current direct detection (DD) experimental bounds while it has a minimal contribution to the observed relic density. The DD cross section of the fermion DM being loop suppressed, resides below the lower limit from the $\{\text{it neutrino floor}\}$, possessing a large fraction of the total relic density.

10. Too Big to Quench? I. Constraining ISM Stripping of Dwarf Satellites in Milky Way-like Halos

Focus: dark matter

Authors: Jingyao Zhu, Stephanie Tonnesen, Greg L. Bryan, Mary E. Putman

Published: 2026-04-08T21:31:03+00:00

PDF: <https://arxiv.org/pdf/2604.07613v1>

Summary: Galaxy environment plays a crucial role in quenching star formation in dwarf galaxies. In Milky Way (MW)-like environments, dwarf satellite quenching is primarily driven by ram pressure stripping (RPS), the direct removal of satellite gas by the host halo gas. Using a suite of 20-pc resolution hydrodynamical wind tunnel simulations, we constrain the satellite mass scale at which the stripping of a dwarf galaxy’s interstellar medium (ISM) becomes inefficient in MW-like halos. The simulations include radiative cooling in a multiphase satellite ISM, star formation, and stellar feedback, and vary both satellite masses ($M_\star = 10^{6.2}, 10^{6.8}, 10^{7.2} M_\odot$) and host halo gas densities along a first-infall and post-pericentric orbit. We find that the degree of ISM stripping in our dwarf galaxies is consistent with the analytical prediction by McCarthy et al. (2008). Star formation is rapidly quenched when RPS is effective, but can be mildly enhanced or temporarily quenched and subsequently reignited when RPS is incomplete. ISM stripping is efficient for satellites with $M_\star \lesssim 10^7 M_\odot$ (or $M_{200} \lesssim 10^{10} M_\odot$) but highly inefficient above this scale. This transitional mass ($M_\star \approx 10^7 M_\odot$) is 0.5-1 dex lower than that found in observations and cosmological simulations, suggesting that additional mechanisms are needed to quench more massive satellites, such as tidal stripping of the satellite dark matter or RPS from a clumpy gaseous halo.

Selected figures

